

A Short Introduction to Machine Learning

Introduction to Machine Learning **Lect. 4**

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Lect 4

Introduction to Machine Learning (continuation)

Introduction to Generalization in ML

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ML in a Nutshell

- **DATA:** available experience represented as vectors, structures,...
 - **TASK:** supervised (classification, regression), unsupervised, ...
 - E.g. Given data as labeled examples, find good approximation of the unknown f .
 - **MODEL:**
 - describes the relationships among the data / the knowledge
 - defines the class of functions that the learning machine can implement (*hypothesis space*)
 - **LEARNING ALGORITHM:**
 - (given data, task and model) the learning algorithm performs a (heuristic) *search* through a space of hypotheses that are valid in the given data
 - E.g. it adapts the free parameters of the model to the task at hand
- **VALIDATION:** evaluate generalization capabilities (of your hp)

ML issues

Easy use of ML tools
versus
correct/good use of ML

ML issues (I)

- Inferring general functions from known data: an *ill posed problem* (e.g. in principle the solution is not unique)
 - With finite data we cannot expect to find the exact solution
- Work with a restricted hypothesis space
 - see also the inductive bias concept
- What can we represent ?
- (Secondary) What can we learn ?
(as if you cannot represent a function you cannot also learn it)

ML issues (II)

Generalization



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- **Learning phase:** to build the model (including training)
- **Prediction phase:** evaluate the learned function over novel samples of data (generalization capability)
- Inductive learning hypothesis
 - Any h that approximates f well on training examples will also approximate f well on new (unseen) instances x (?)

Def

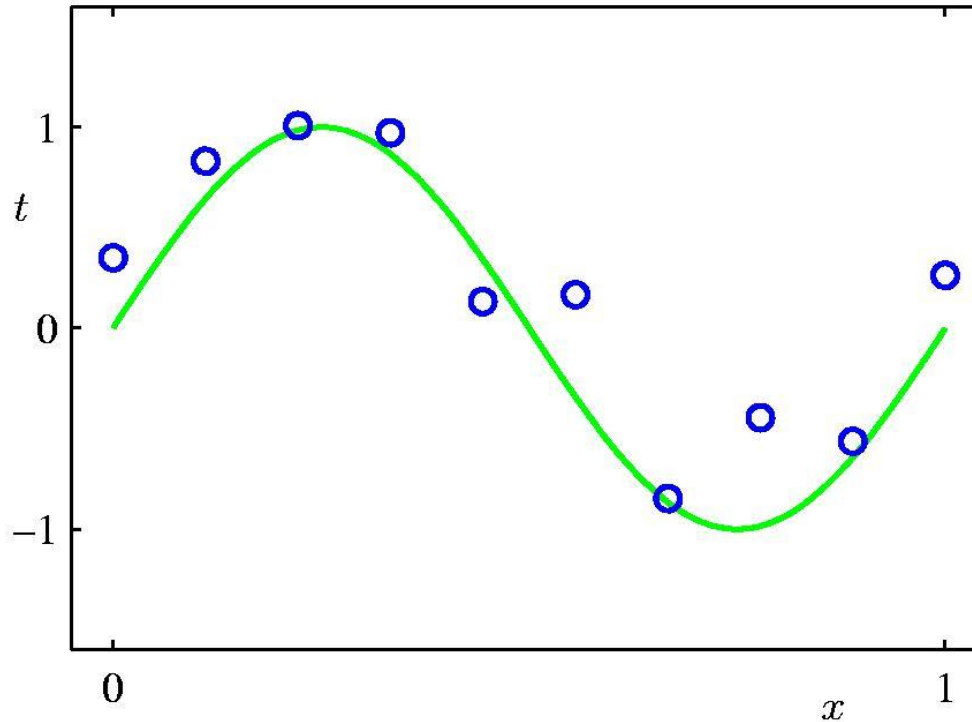
- **Overfitting:** A learner overfits the data if
 - it outputs a hypothesis $h(\cdot) \in H$ having true/generalization error (risk) R and empirical (training) error E , but there is another $h'(\cdot) \in H$ having $E' < E$ and $R' < R$ (so that $h'(\cdot)$ is the better one, despite a worse fitting).
- Critical aspect: accuracy / performance estimation
 - Theoretical
 - Empirical (training, test) and cross-validation techniques

Complexity on a case study

- An example on a parametric model for *regression*:
- The set of functions is assumed as polynomials with degree M
- The **complexity** of the hypothesis increases with the degree M
- l = number of examples
- Warning: This is an artificial simplified task (unrealistic due to the use of just 1 input variable, the fact that we know the target function in advance, ...)

Polynomial Curve Fitting

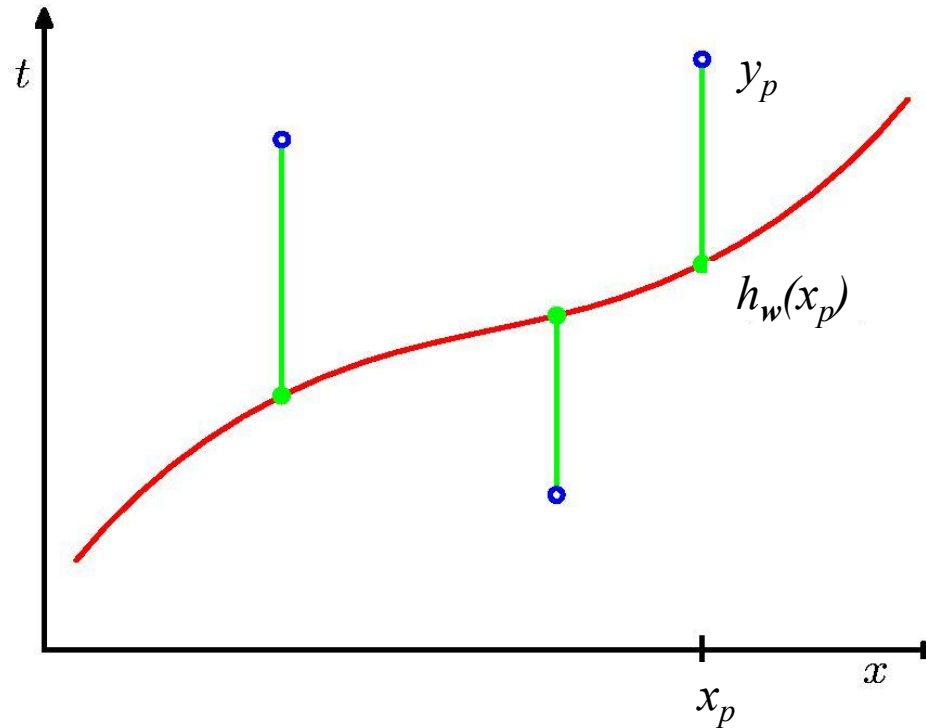
Target = $\sin(2\pi x)$ + random noise (gaussian)



$$h_{\mathbf{w}}(x) = w_0 + w_1x + w_2x^2 + \dots + w_Mx^M = \sum_{j=0}^M w_jx^j$$

Samples affected by noise (not always on the green “true” line)

Sum-of-Squares Error Function

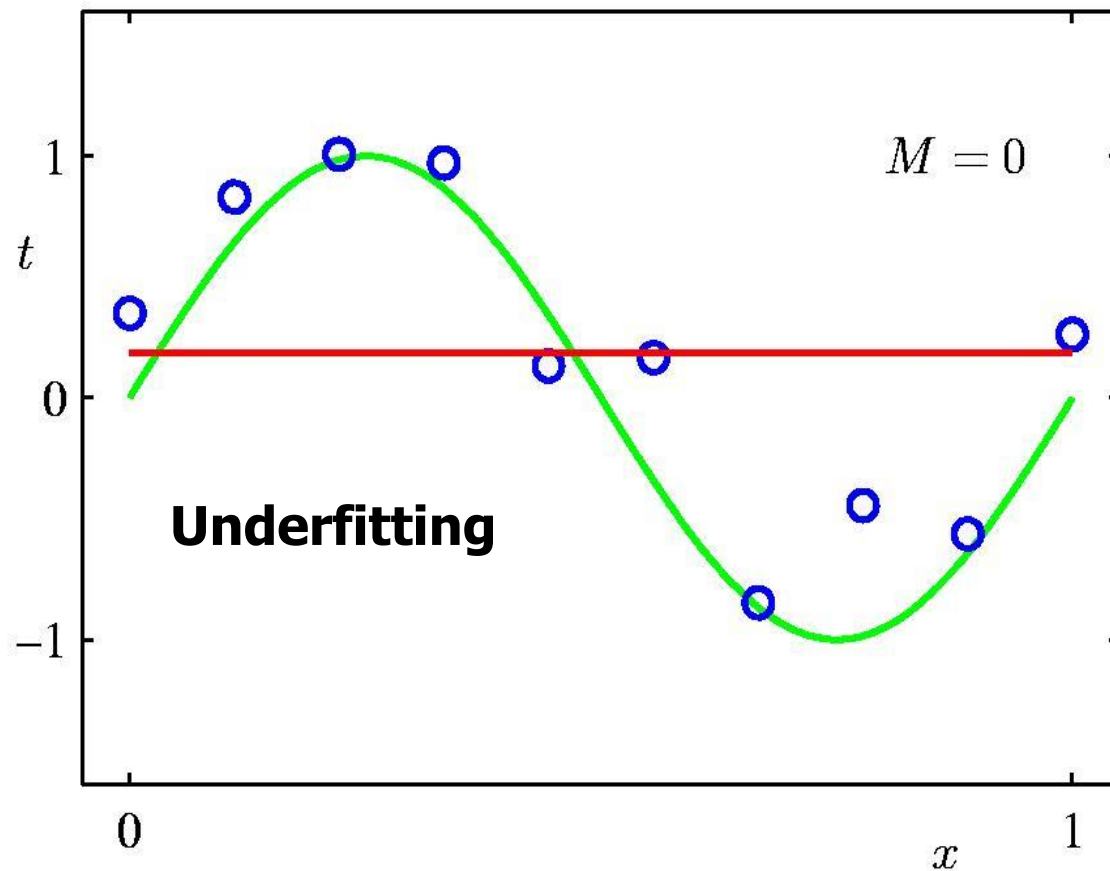


$$E(\mathbf{w}) = \sum_{p=1}^l (y_p - h_{\mathbf{w}}(x_p))^2$$

Note: p is the example, y_p the target for p
 l the total number of examples
 $h_{\mathbf{w}}(x_p)$ is the model output at the point x_p
 $(x$ is a single variable, $n=1)$

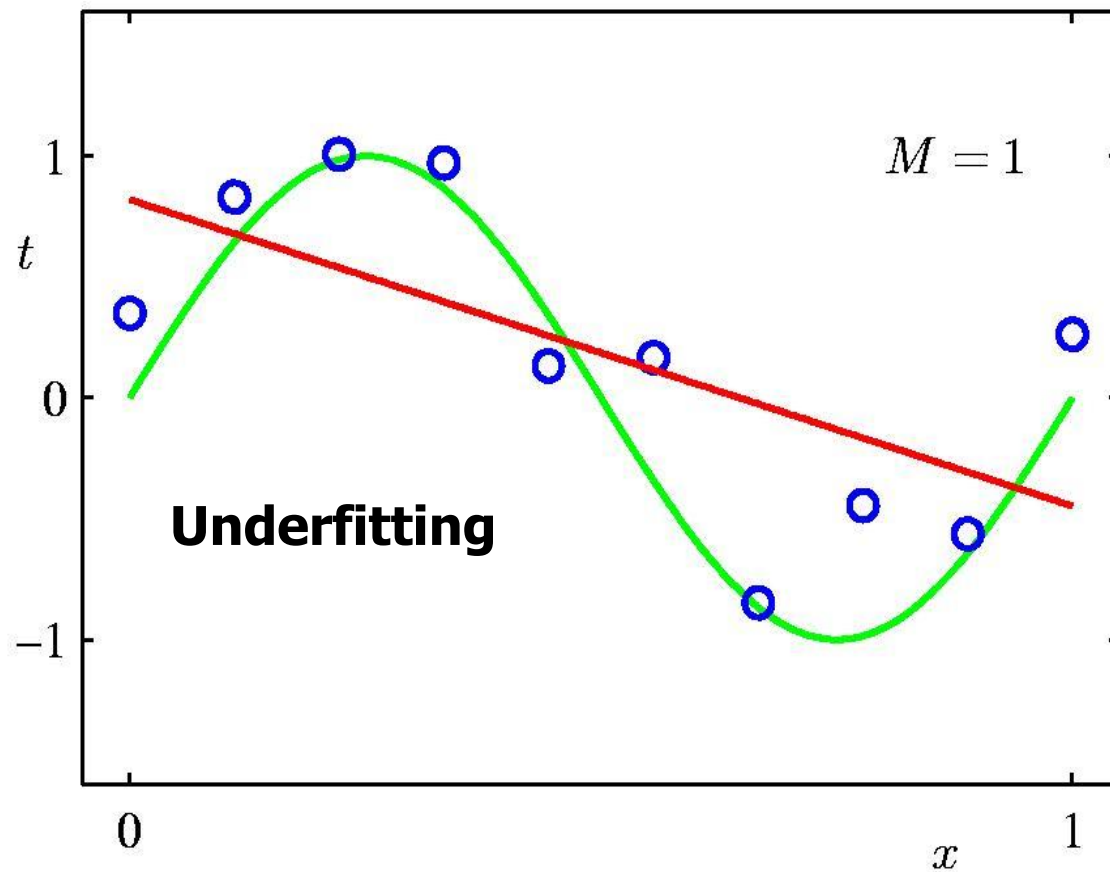
Minimize $E(\mathbf{w})$ (Square Error) to find the best $\mathbf{w}$ (fitting)

0th Order Polynomial



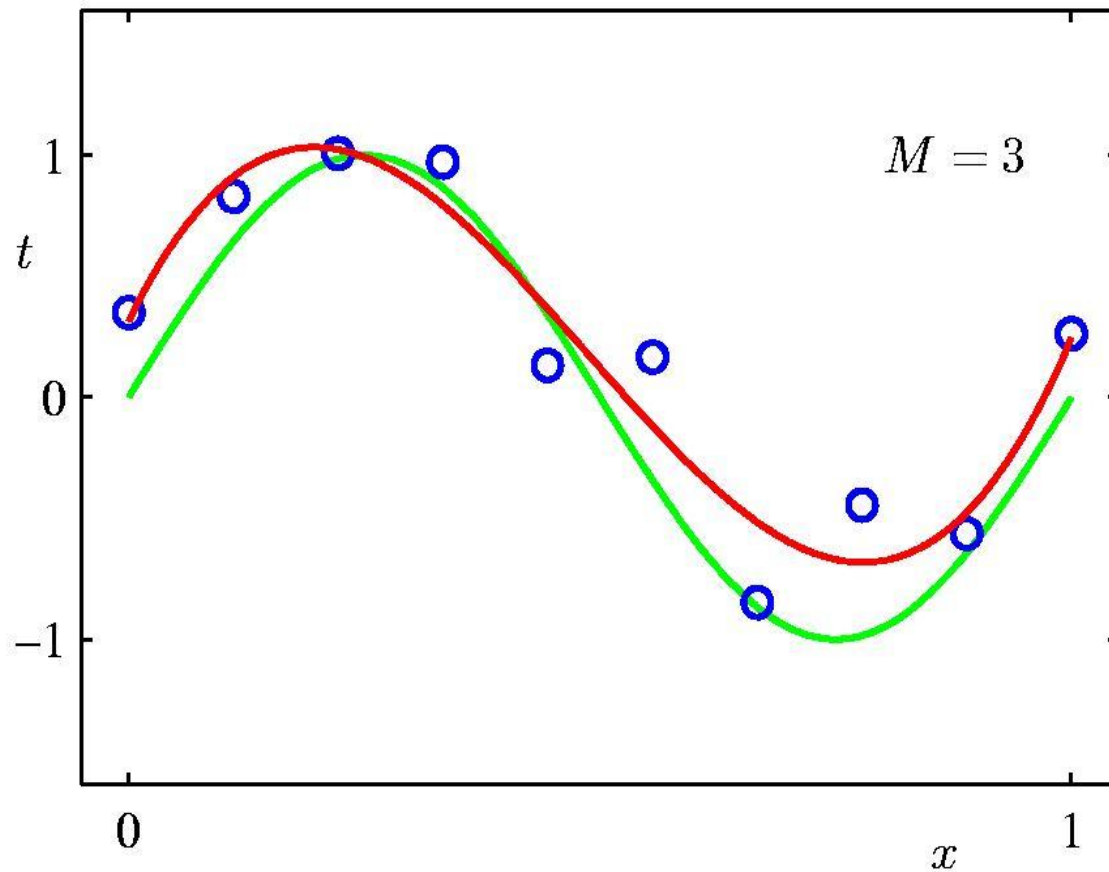
Underfitting: too simple model (red line)
w.r.t. to the target function

1st Order Polynomial



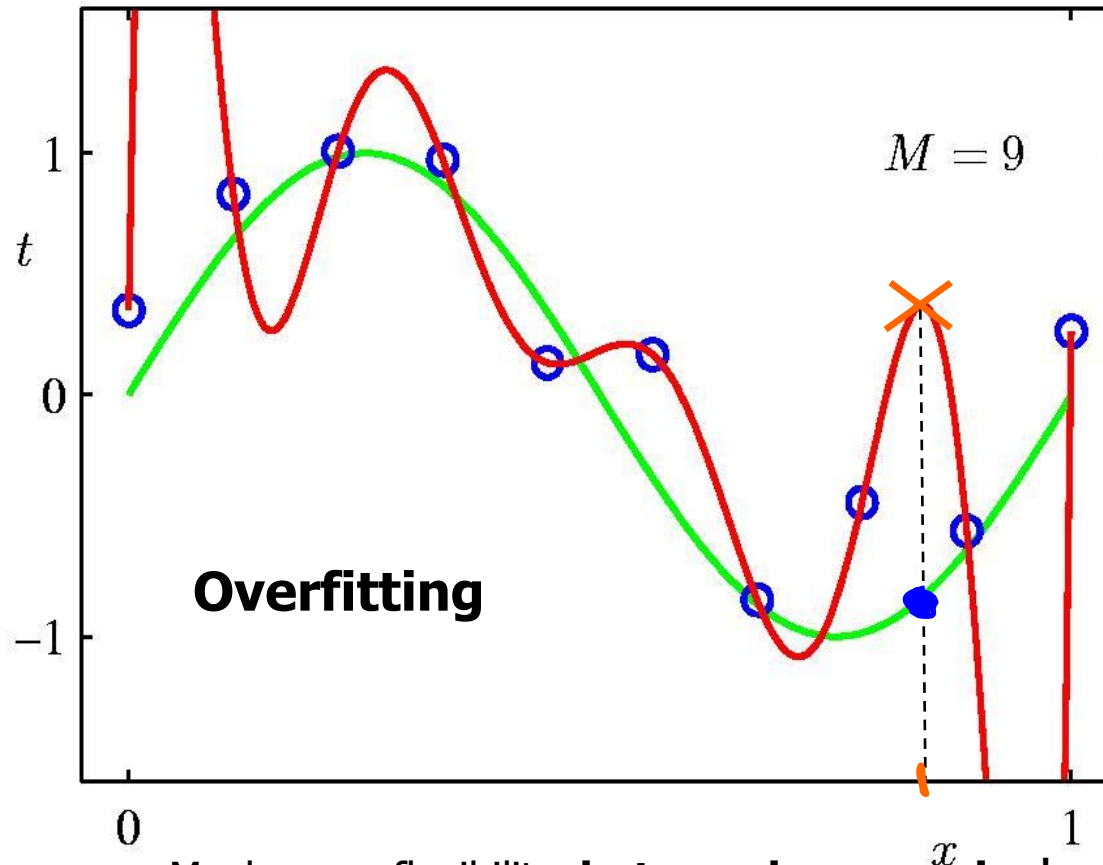
Still poor solution (due to **underfitting**)

3rd Order Polynomial



More **flexibility** is useful !!!

9th Order Polynomial



Much more flexibility, **but may be excessive!**

$E(\mathbf{w}) = 0$ on training data!!! But error on test set ?

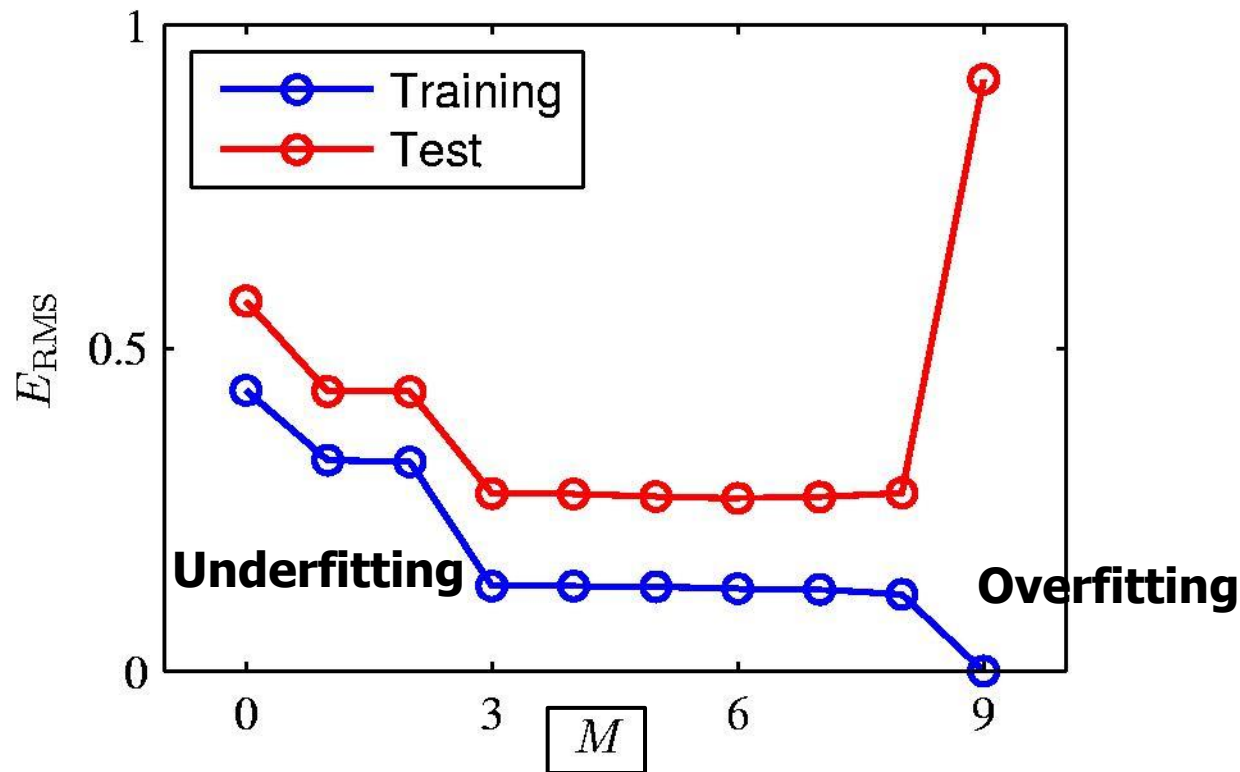
Too complex model (in this case it fits even the noise)!

Poor representation of the (green) true function, due to **overfitting**

Underfitting and Overfitting with the complexity (M)



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Root-Mean-Square (RMS) Error: $E_{\text{RMS}} = \sqrt{2E(\mathbf{w}^*) / l}$

Where $E(\mathbf{w}^*)$ is the error for the trained model

Polynomial Coefficients

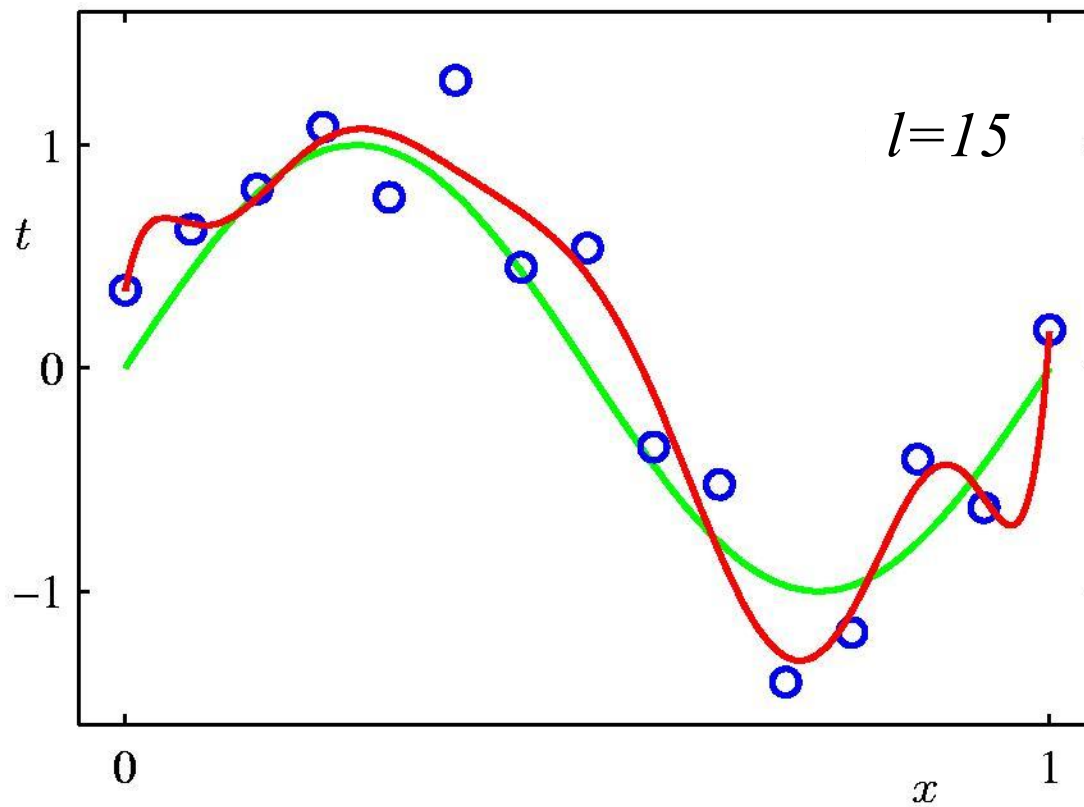
	$M = 0$	$M = 1$	$M = 3$	$M = 9$
w_0^*	0.19	0.82	0.31	0.35
w_1^*		-1.27	7.99	232.37
w_2^*			-25.43	-5321.83
w_3^*			17.37	48568.31
w_4^*				-231639.30
w_5^*				640042.26
w_6^*				-1061800.52
w_7^*				1042400.18
w_8^*				-557682.99
w_9^*				125201.43

Data Set Size: $l=15$
previous was 10



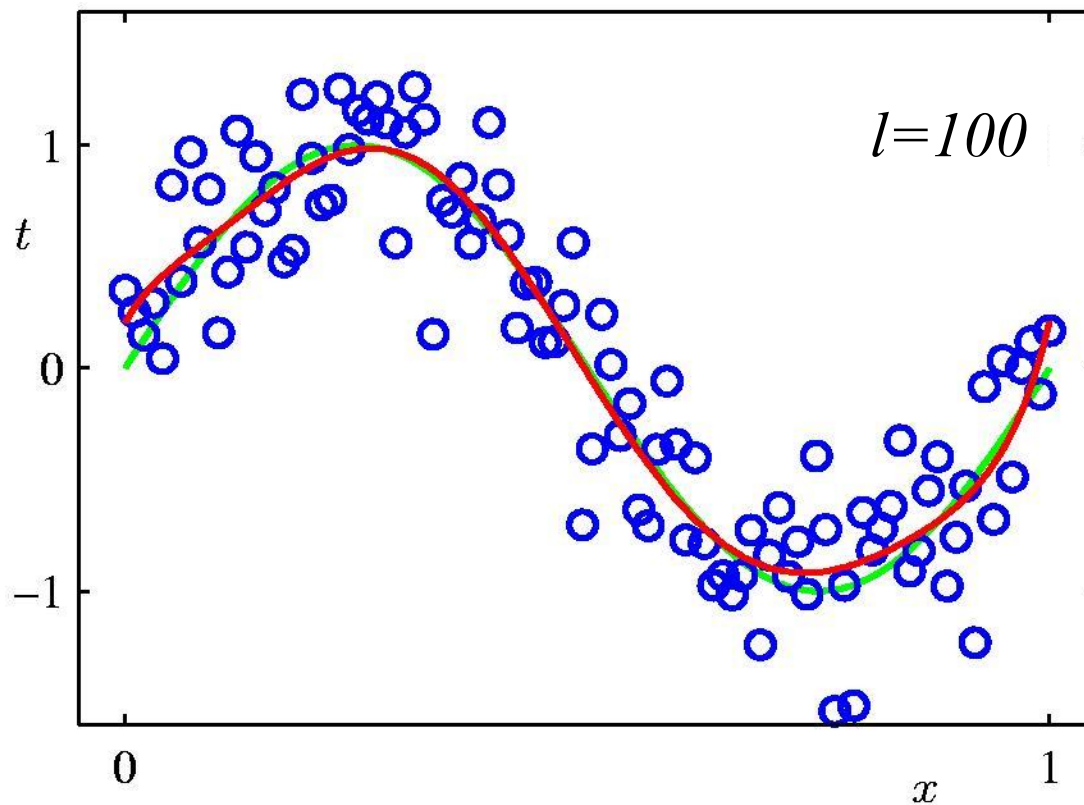
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9th Order Polynomial



Data Set Size: $l=100$

9th Order Polynomial (even more data)



We can use higher M with a higher number of data

Toward SLT

Putting all together:

- We want to investigate on the **generalization** capability of a model (measured as a risk or test error)
 - with respect to the **training error**
 - **overfitting** and **underfitting** zones
- The role of model complexity
- The role of the number of data
- ***Statistical Learning Theory (SLT)***: a general theory relating such topics

(Simplified) Formal Setting

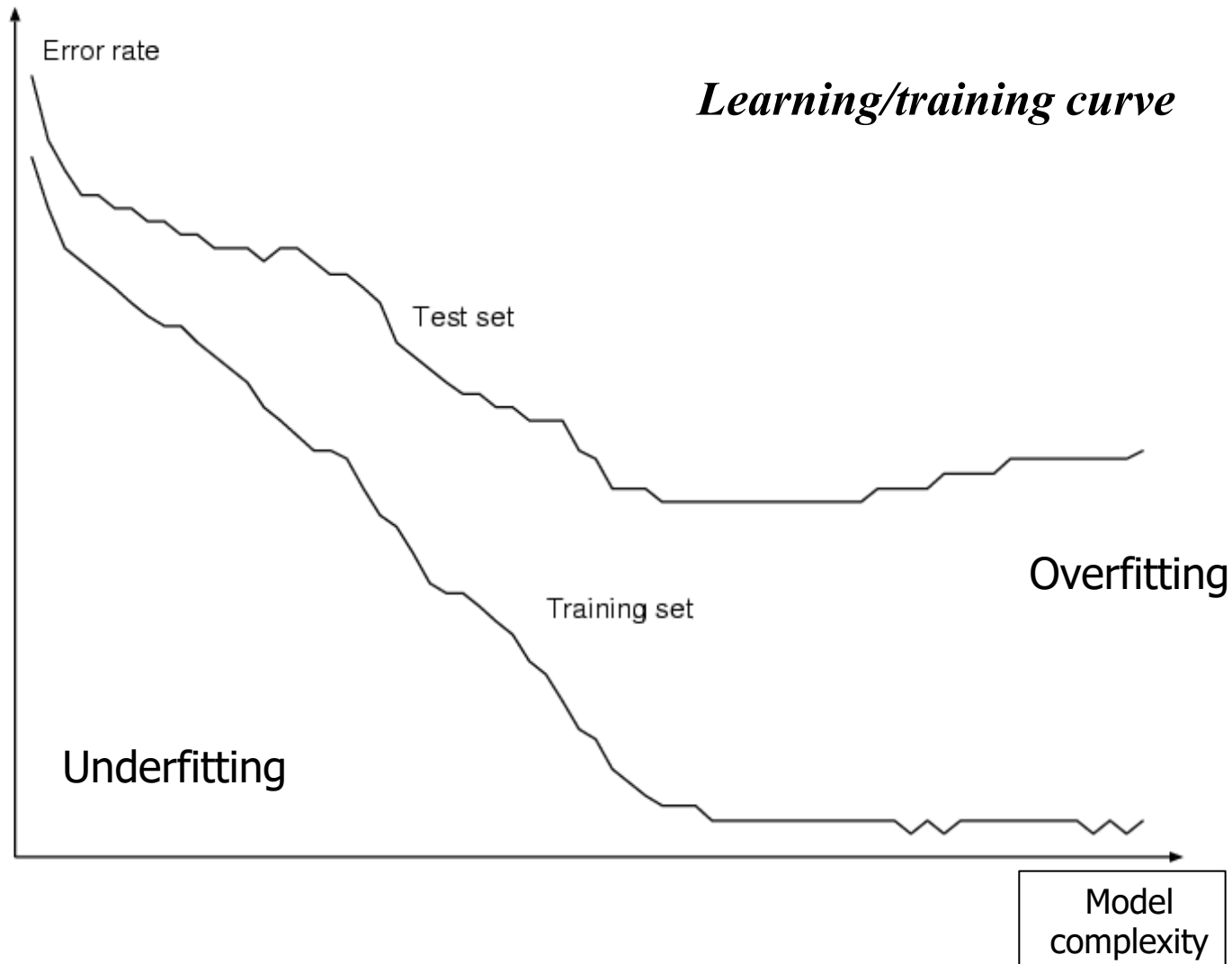
- Approximate unknown $f(\mathbf{x})$, d is the target ($d = \text{true } f + \text{noise}$)
- Minimize *risk function* $R = \int L(d, h(\mathbf{x})) dP(\mathbf{x}, d)$ True Error
Over *all* the data
- Given
 - value from teacher (d) and the probability distribution $P(\mathbf{x}, d)$
 - a loss (or cost) function, e.g. $L(h(\mathbf{x}), d) = (d - h(\mathbf{x}))^2$

- Search h in H : Min R
- But we have only the finite data set $TR = (\mathbf{x}_p, d_p), \quad p = 1 \dots l$
- To search h : minimize empirical risk (training error E), finding the best values for the model free parameters

$$R_{emp} = R_{emp}(h, TR) = \frac{1}{l} \sum_{p=1}^l (d_p - h(\mathbf{x}_p))^2$$

- Empirical Risk Minimization (ERM) Inductive Principle
- *Can we use R_{emp} to approximate R ?*

Typical behavior of learning



Vapnik-Chervonenkis-dim and SLT: a general theory (I)



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- Given the *VC-dim* (VC), a measure *complexity* of H (*flexibility to fit data*) (e.g. Num. of parameters for linear models/polynomials)

Repetita: Can we use R_{emp} to approximate R ?

Very
important!

Def.

VC-bounds in the form: it holds with probability $1-\delta$ that
guaranteed risk

$$R \leq \underbrace{R_{emp}} + \underbrace{\varepsilon(1/l, VC, 1/\delta)}_{VC\text{-confidence}}$$

- First (basic) explanation, ingredients:
 - ε is a function that grows with VC ($VC\text{-dim}$), and that decreases with (higher) l and δ (i.e. l and δ are in the denominator)
 - We know that R_{emp} decreases using complex models (with high $VC\text{-dim}$) (e.g. the polynomial degree in the example)
 - δ is the confidence, it rules the probability that the bound holds (e.g. low delta 0.01 $\rightarrow$ the bound holds with probability 0.99)
- Now we can see how it can “explain” the *underfitting* and *overfitting* and the aspects that control them.

Vapnik-Chervonenkis-dim and SLT: a general theory (II)



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Comments:

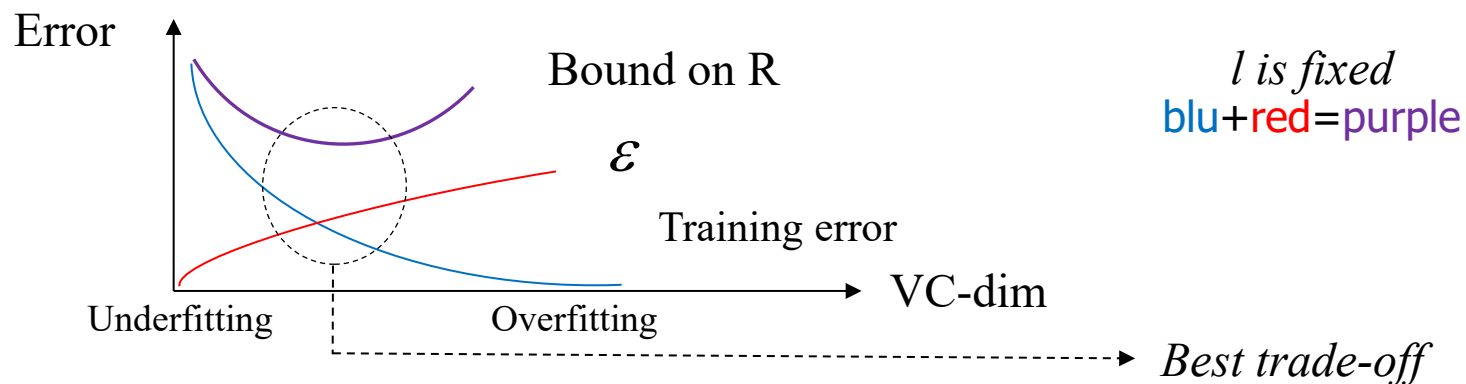
- *VC-bounds in the form:* it holds with probability $1-\delta$ that

$$\text{guaranteed risk } R \leq \underbrace{R_{emp}}_{\text{Training error}} + \underbrace{\varepsilon(1/l, VC, 1/\delta)}_{\text{VC-confidence}}$$

Very
important!

Intuition:

- Higher l (data) $\rightarrow$ lower *VC-confidence* and a bound close to R
- Too simple models (low VC-dim) can be not suff. due to high R_{emp} (underfitting)
- Higher VC-dim (fix l) $\rightarrow$ lower R_{emp} but *VC-conf.*, and hence R , may increase (overfitting)
- Structural risk minimization: minimize the bound !



- Concept of control of the model complexity (flexibility):
trade-off between model complexity (VC-dim) and TR accuracy (fitting)

An example

It is possible to derive an upper bound of the ideal error which is valid with probability $(1-\delta)$, δ being arbitrarily small, of the form:

- General:
$$R \leq R_{emp} + \varepsilon(1/l, VC, 1/\delta)$$
- Example:
$$R \leq R_{emp} + \varepsilon(VC/l, -\ln(\delta/l))$$
- There are different bounds formulations according to different classes of f , of tasks, etc.
- More in general, in other words (simplifying): we can make a good approximation of f from examples, provided we have a good number of data, and the complexity of the model is suitable for the task at hand.
 - Fit data as much as possible to avoid underfitting (high R_{emp}), but not too much so to avoid overfitting (due to the increase of *VC-confidence* term)
 - (Double descent/over-parametrization/benign overfit phenomena will be discussed later)

Discussion

Complexity control



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- SLT - Statistical Learning Theory:
 - It allows formal framing of the problem of generalization and overfitting, providing analytic upper-bound to the risk R for the prediction over all the data, regardless to the type of learning algorithm or details of the model
 - *The ML is well founded*: the Learning risk can be analytically limited and only few concepts are fundamentals !
 - It leads to new models (SVM) (and other methods that directly consider the control of the complexity in the construction of the model)
 - It bases one of the inductive principles on the control of the complexity
 - It explains the main difference with respect to supporting methods from CM (providing the techniques to perform fitting), apart from modelling aspects

Open questions:

- What (other) principles are to found the control of the complexity? How to work in practice?
 - How to measure the complexity (or fitting flexibility)?
 - How find the best trade-off between fitting and model complexity?

Exercises

- Reinterpretation of some parts in the first lectures: Why a zero error for the training does not necessarily imply a good solution?
- Connect again by your-self the underfitting and overfitting to the SLT inequality interpretation
- Connect again by your-self to the example with polynomials
- Looking at the def. of overfitting, denote the h and h' on the plot of the SLT bound.
- Is the presence of noise the cause of overfitting (or the complexity trade-off)? Can you image an example, like the overfitting with a polynomial with high degree, were even with completely cleaned data (no noise) you have overfitting?

Validation

- Evaluation of performances for ML systems =
Generalization/Predictive accuracy evaluation
- “*The performance on training data provide an overoptimistic evaluation*”
- Validation !
- Validation !!
- Validation !!!
- In the following: an introduction, → 
- Validation will be the topic of specific lectures later, in the “*Validation & SLT*” part of the course
- And it has a *central role* for the applications and the project

Very
important!

Validation: Two aims

- After models training on the training set
- **Model selection:** estimating the performance (*generalization error*) of different learning models in order to choose the best one (to generalize).
 - this includes search the best *hyper-parameters* of your model (e.g. polynomial order, ...).

It returns a model

- **Model assessment:** having chosen a final model, estimating/evaluating its prediction error/ risk (*generalization error*) on new test data (measure of the quality/performance of the ultimately chosen model).

It returns an estimation

Gold rule: Keep separation between goals and use separate data sets

Validation: ideal world

- A large *training set* (to find the best hypothesis, see the theory)
- A large validation set for model selection
- A very large external unseen data *test set*

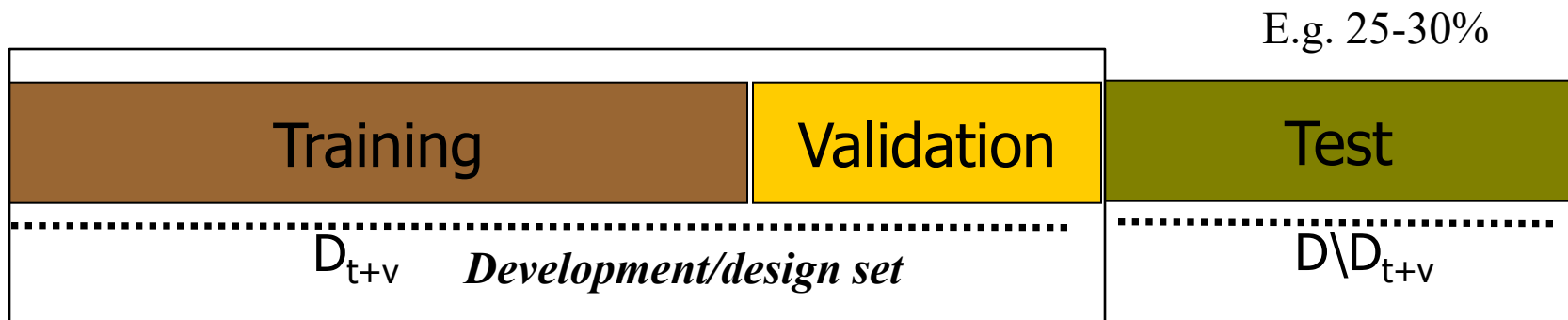
- *With finite and often small data sets?*
- *....Just estimation of the generalization performance*

- *We anticipate to basic techniques:*
 - *Simple hold-out (basic setting)*
 - *K-fold Cross Validation (just an hint in this lecture)*

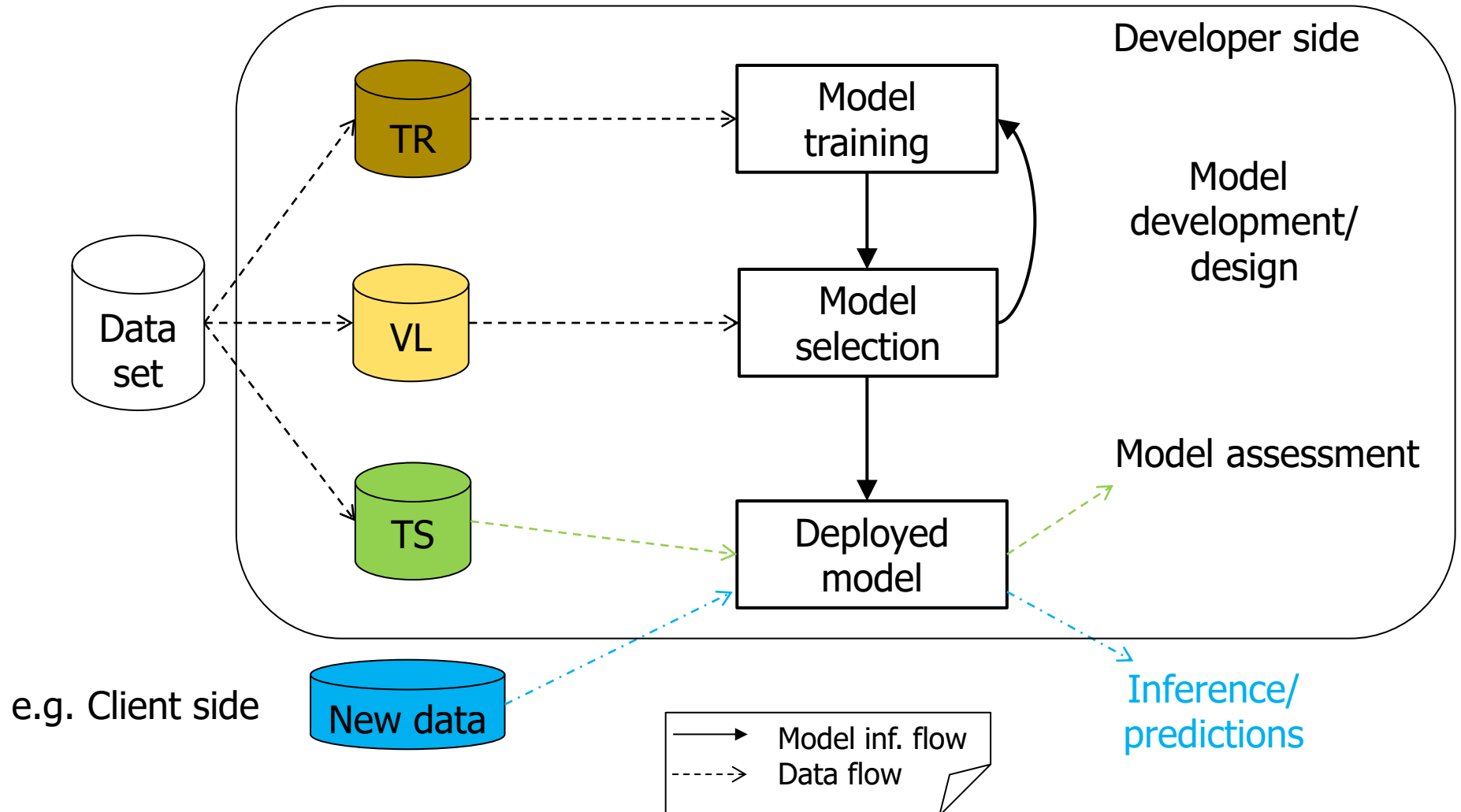
Hold out cross validation

Hold out: basic setting

- Partition data set D into *training set* (TR), *validation or selection set* (VL) and *test set* (TS)
 - All the three sets are disjoint sets !!!
 - **TR** is used to run the **training** algorithm
 - **VL** can be used to **select the best model** (e.g hyper-parameters tuning)
 - **Test** set (result) is *not* to be used for tuning/selecting the best h : it is only for **model assessment**



TR/VL/TS by a schema



Hold out and **K-fold** cross validation



Hold out CV *can make insufficient use of data*



K-fold Cross-Validation (we will see later in a specific lecture)

- Split the data set D into k mutually exclusive subsets $D_1, D_2, \dots, D_K$
- Train the learning algorithm on $D \setminus D_i$ and test it on D_i
- Can be applied for both VL or TS splitting
- *It uses all the data for training and validation or testing*

Issues:

- How many folds? 3-fold, 5-fold, 10-fold, ..., 1-leave-out
- Often computationally very expensive
- Combinable with validation set, double-K-fold CV,

Classification Accuracy

Def

Confusion matrix

Predicted \ Actual	Positive	Negative
Positive	TP	FN
Negative	FP	TN

$$\text{Specificity} = \text{TN} / (\text{FP} + \text{TN})$$

(*True Negative rate = 1 - FPR*)

$$\text{Sensitivity} = \text{TP} / (\text{TP} + \text{FN})$$

(*True Positive rate or Recall*)

$$(\text{Precision} = \text{TP} / (\text{TP} + \text{FP}))$$

false positive (FP) :eqv. with
false alarm

Accuracy: % of correctly classified patterns = $\text{TP} + \text{TN} / \text{total}$

Note: for binary classif.: 50% correctly classified = “coin” (random guess) predictor!

Other topics:

- unbalanced data (e.g. 99% +) → trivial classifier exists ,
-

ROC curve

Confusion matrix

Actual \ Predicted	Positive	Negative
Positive	TP	FN
Negative	FP	TN

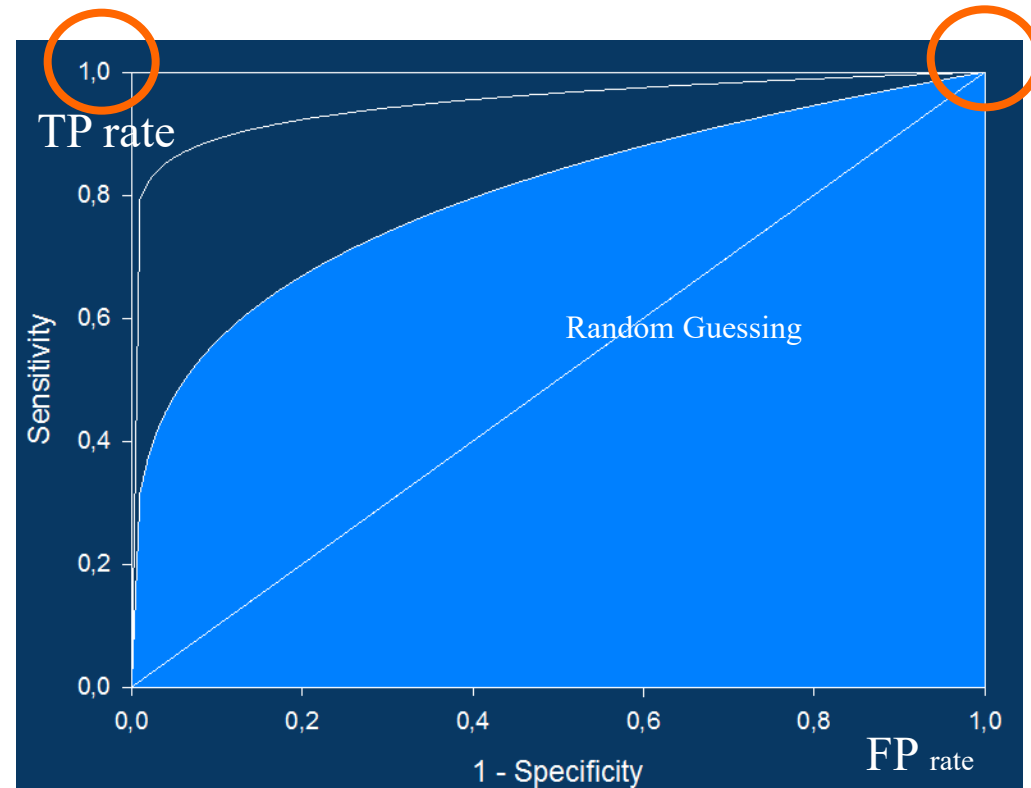
$$\text{Specificity} = \text{TN} / (\text{FP} + \text{TN})$$

$$\text{TPr or Sensitivity} = \text{TP} / (\text{TP} + \text{FN})$$

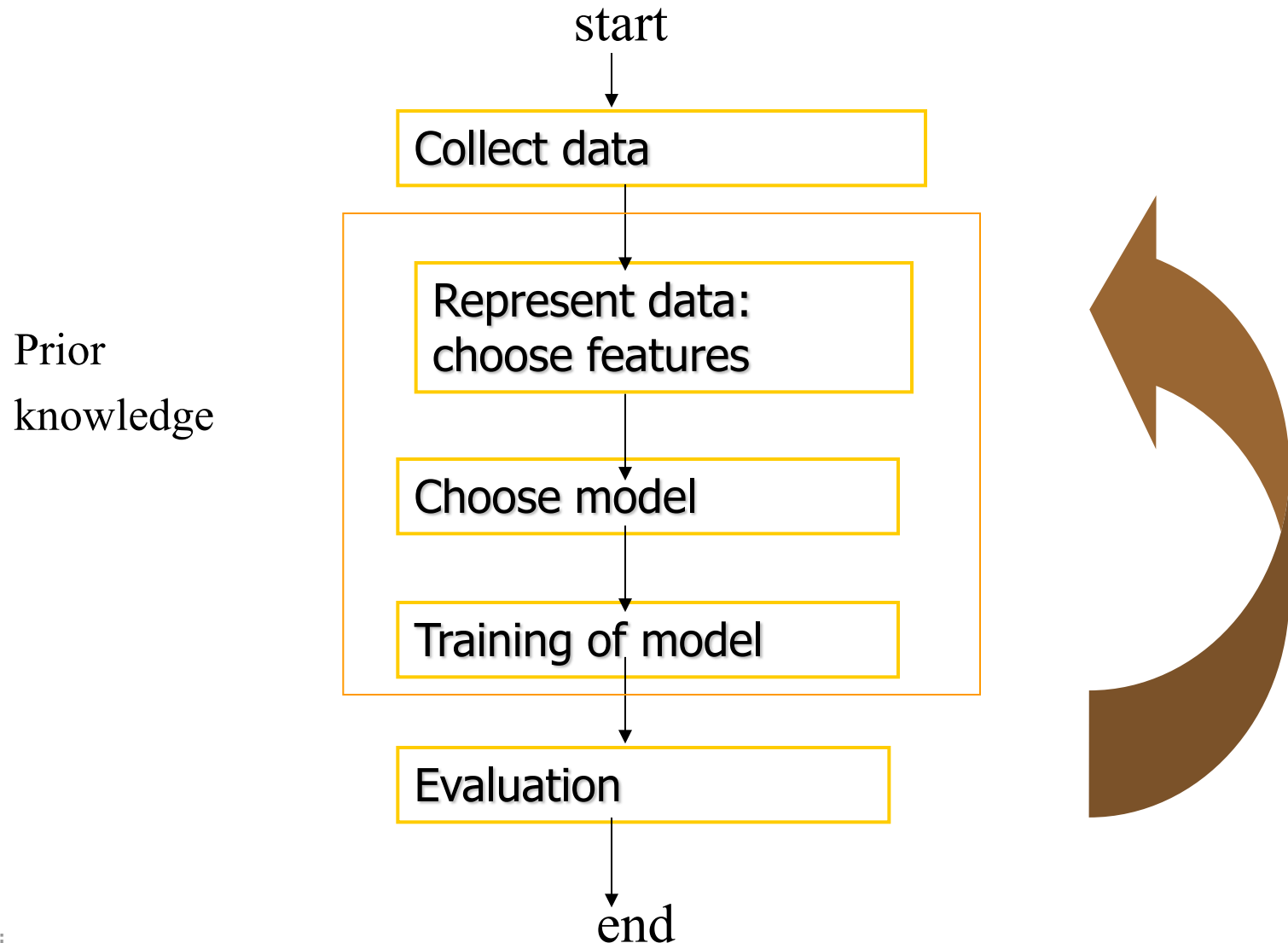
ROC curve

The diagonal corresponds to the worst classifier.

- Better curves have higher AUC (Area Under the Curve).



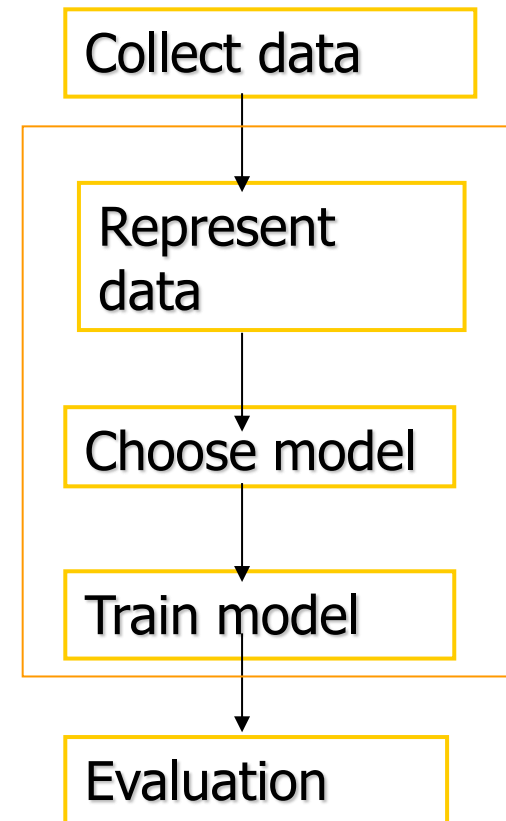
The Design Cycle



Design cycle (I)

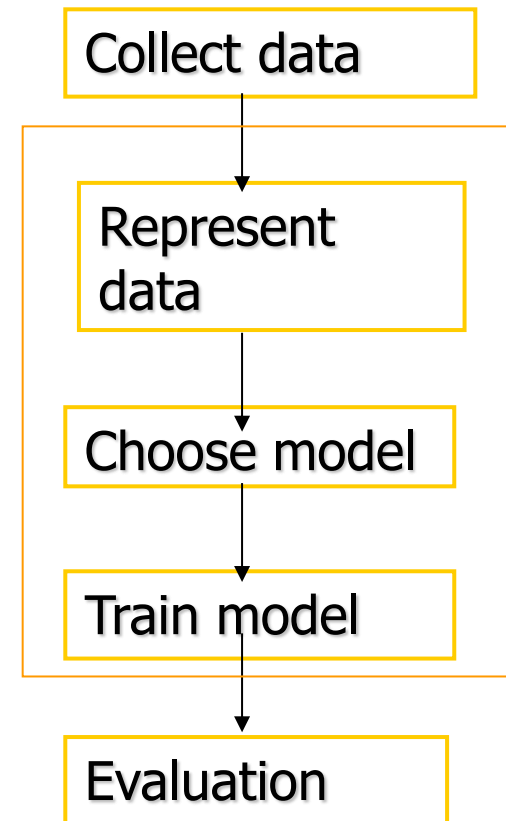
- **Data collection:**
 - Selection, integration, data cleaning etc.
 - Adequately large and representative set of examples for training and test
- **Data representation**
 - Domain dependent, exploit prior knowledge of the application expert
 - Feature selection
 - Outliers detection
 - Other preprocessing: variable scaling, missing data,..

Often the most critical phase for an overall success!
- **Model choice:**
 - Statement of the problem
 - Hypothesis formulation
 - You must know the limits of applicability of your model
 - complexity control



Design cycle (II)

- Building of the model (core of ML):
 - through the learning algorithm using the training data
- Evaluation:
 - Performance = predictive accuracy !
 - Also *interpretation* of model outcomes and *explanation* of the results
 - “knowledge” extraction
- Deployment...



Misinterpretations

For every statistical models (including DM applications)

- Causality is (often) assumed and a set of data representative of the phenomena is needed.
 - Not for unrelated variables and for random phenomena (lotteries)
 - Uninformative input variables → poor modeling → Poor learning results
- Causality cannot be inferred from data analysis alone:
 - People in Florida are older(on av.) than in other US states.
 - Florida climate causes people to live longer ?
- May be there is a statistical dependencies for reasons outside the data
- More specifically for ML:
- Powerful models (even for “garbage” data) → higher risk !
- Not-well validated results: the predicted outcome and the interpretation can be misleading.

Bibliographic references (lect 1-2-3-4)

- **Course notes (slides copy): lectures 1-4** without specific textbook materials
 - *Readings* : Mitchell. *The discipline of Machine Learning*. July 2006. CMU-ML-06-108
 - And other in: <http://www.di.unipi.it/~micheli/DID/>
- *On the textbook:*
 - S. **Haykin**: *Neural Networks: a comprehensive foundation*, IEEE Press, 1998. (2nd. Ed.): **sez.1.7** (knowledge rep)
 - **(3rd ed): sez.1.7** (knowledge rep), **1.8, 1.9** (learning processes and tasks)
 - T. M. **Mitchell**, *Machine learning*, McGraw-Hill, 1997: **cap 1 and 2**.
- *Other references:*
 - Russell, Norvig: *Intelligenza artificiale (AIMA)*, 2005 (in vol. 2)
 - E.g. background appendix: <http://aima.cs.berkeley.edu/newchapa.pdf>
 - Hastie, Tibshirani, Friedman, *The Elements of Statistical Learning*, Springer Verlag, 2001 (esiste New Ed.): **cap 1** e sez. 7.10
 - Cherkassky, Mulier, *Learning from data : concepts, theory, and methods*, Wiley, 1998 (esiste New Ed.): **cap1 e sez.2.1** (loss e tasks)
 - C.M. Bishop, *Pattern Recognition and Machine Learning*, Springer 2006: **sez. 1.1** (polynomial fitting example)
 - Duda, Hart, Stork, *Pattern Classification*, 2nd. ed. J. Wiley & Sons, 2001: **cap1** (design cycle)

ML Course structure

Where we go



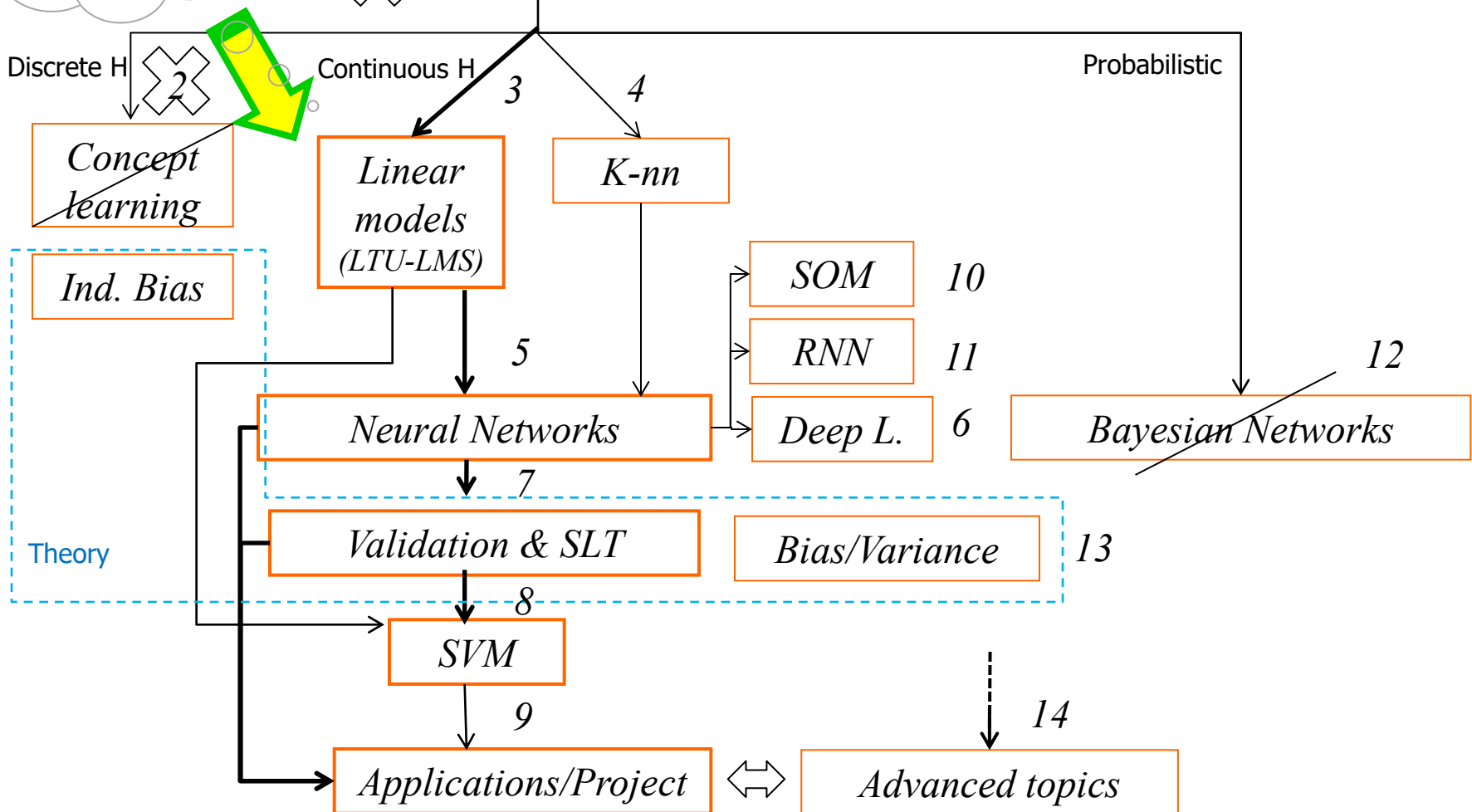
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Firsts learning
alg on a simple
hp space



INTRO

Function approximation framework
Data, Task, Model, Learn. Alg., Validation



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